

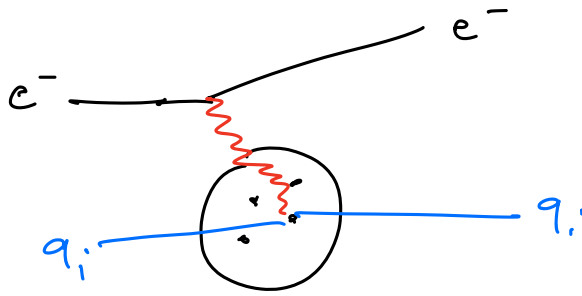
Quark Parton Model

$e^- + p \rightarrow e^- + X$ inelastic

$$\Rightarrow e^- + p \Rightarrow e^- + q \rightarrow e^- + X$$

q : partons

$e^- + X$: incoherent elastic scattering on partons



$$\left. \frac{d\sigma}{d\Omega} \right|_{DIS} \approx \frac{\alpha^2}{q^4} E'^2 \left[W_2(Q^2, \nu) \cos^2 \frac{\theta}{2} + 2 W_1(Q^2, \nu) \sin^2 \frac{\theta}{2} \right]$$

introduced: parton density functions.

$$F_2(Q^2, \nu) = \nu W_2(Q^2, \nu)$$

$$F_1(Q^2, \nu) = M W_1(Q^2, \nu)$$

$$F_2(Q^2, \nu) = 2x F_1(Q^2, \nu)$$

measure $\frac{d\sigma}{d\Omega} \Rightarrow$ determine $\bar{F}_1, \bar{F}_2$

F_1, F_2 do not depend strongly on Q^2

$$Q^2 = M^2 - W^2 + 2M\nu$$

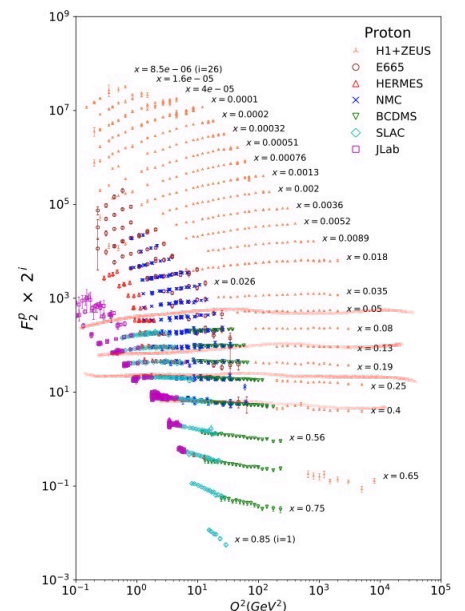
$$Q^2 = 4EE' \sin^2 \frac{\theta}{2}$$

$$\nu = E - E'$$

E, E', θ measured from the e^-

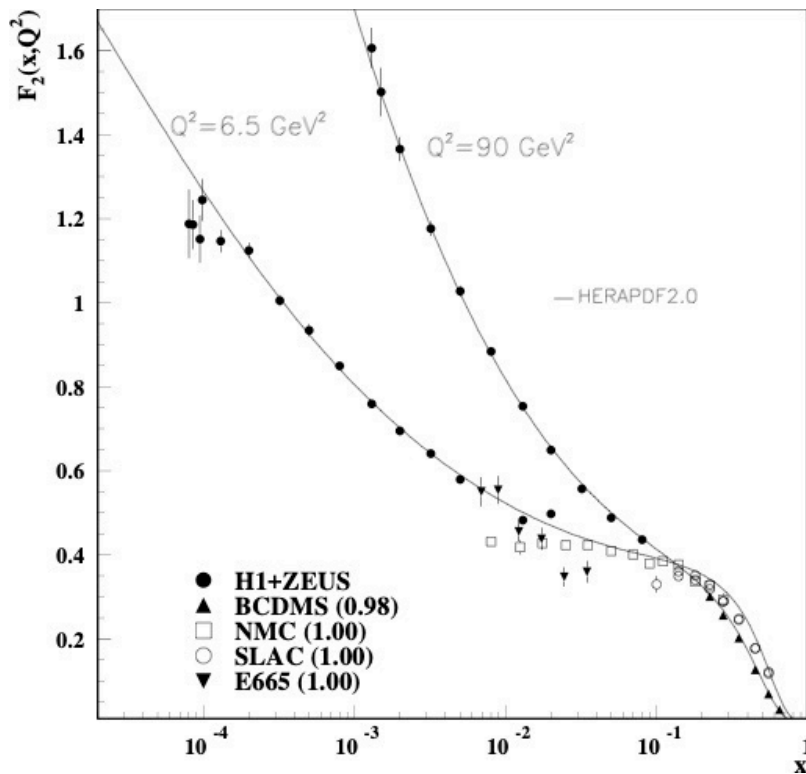
$$x = \frac{Q^2}{2M\nu}$$

M : mass of the target
 $M \equiv M_{\text{proton}}$



Björken scaling: F_1, F_2 do not depend on Q^2, ν separately

$\Rightarrow$ only depend on some ratio.



$$\Rightarrow F_2(Q^2, \nu) \equiv F_2(x) \quad F_1(Q^2, \nu) \equiv F_1(x)$$

$$\int_0^1 f(x) dx = 1 \quad \text{parton density function}$$

$$F_2 = \frac{1}{2} \sum_i e_q^2 f_i(x) \quad i: \text{sum over partons.}$$

$$x_B = \frac{Q^2}{2M\nu} \quad \text{determined from kinematics of } e^-$$

$$x = \frac{m}{M} \quad \begin{array}{l} m: \text{mass proton} \\ M: \text{mass of target/proton} \end{array}$$

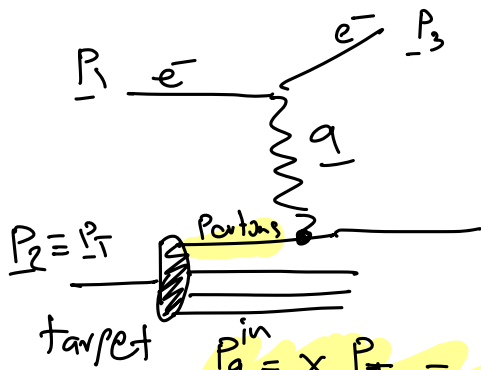
Feynman Interpretation of x

$$| \text{Diagram 1} |^2 = | \text{Diagram 2} |^2 + | \text{Diagram 3} |^2 + \dots$$

Diagram 1: A horizontal line representing a proton with a vertical wavy line (photon) hitting it at a point. The incoming electron momentum is $\underline{P}_2 = \underline{P}_T$.

Diagram 2: A horizontal line representing a proton with a vertical wavy line (photon) hitting it at a point, and a small circle (representing a parton) attached to the line.

Diagram 3: A horizontal line representing a proton with a vertical wavy line (photon) hitting it at a point, and a larger circle (representing a parton) attached to the line.



$$E = 25 \text{ GeV}$$

$$\underline{P}_q^{\text{fin}} = \underline{P}_q^{\text{in}} + \underline{q} : \text{parton}$$

$$\underline{P}_A = \underline{P}_2 + \underline{q} : \text{proton}$$

$$\underline{P}_q^{\text{in}} = x_F \underline{P}_T = x_F (M, 0, 0, 0)$$

e^- interacts with 1 parton
the other partons are spectators

$$\vec{P}_L$$

$$e^- \longrightarrow \longleftarrow q$$

in CMS,

$$E \approx 25 \text{ GeV}$$

longitudinal momentum

transverse momentum of partons $|\vec{P}_T| \ll |\vec{P}_L|$
in high energy regime.

$$x_F = \frac{|\vec{P}_L|}{|\vec{P}_L|}$$

fraction of momentum carried by partons.

$$|\underline{P}_q^{\text{fin}}|^2 = \underbrace{|\underline{P}_q^{\text{in}}|^2}_{m_q^2} + |\underline{q}|^2 + 2 \underline{P}_q^{\text{in}} \cdot \underline{q} = -Q^2 + 2 x_F M V$$

$$\underline{P}_q^{\text{in}} = x_F (M, 0, 0, 0)$$

$$\underline{q} = (\underbrace{E-E'}_V, \underbrace{\vec{P}_1 - \vec{P}_3}_q)$$

$m_q^2 \ll Q^2$: negligible mass for partons.

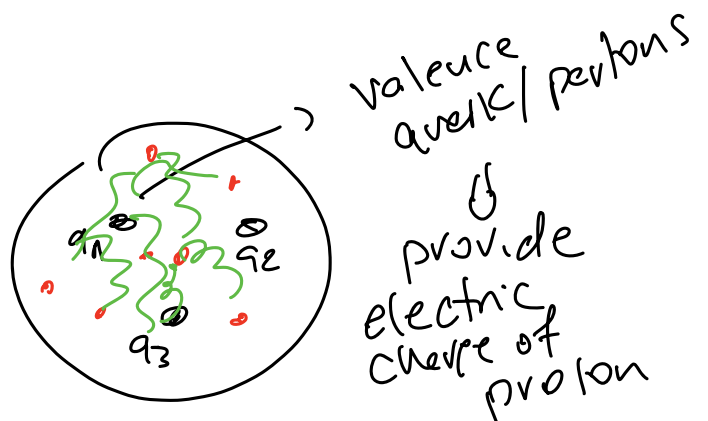
$$\Rightarrow Q^2 = 2 x_F M V \Rightarrow x_F = \frac{Q^2}{2 M V}$$

$$\Sigma: \underline{P}_q^{\text{in}} = \underline{P}_e \approx \underline{P}_{\text{Target}}$$

in high energy limit
 $E_e \approx 25 \text{ GeV} \gg m_e, m_q$

$$x_F \equiv x_B = x$$

How many partons?



proton: $q_1 + q_2 + q_3 = +1$

$P_{q_1} + P_{q_2} + P_{q_3} = P_{\text{proton}}$

Experimental evidence $\sum_i^{\text{valence}} x_i \approx 0.5 < 1$

$\Rightarrow$ valence quarks not sufficient to conserve momentum

$\Rightarrow$ sea quarks:

proton: uud

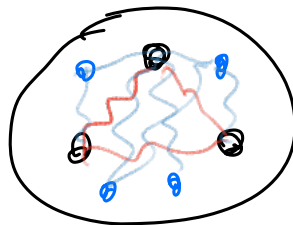
$$Q = +\frac{2}{3} \begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix} - \frac{1}{3} \begin{pmatrix} \bar{d} \\ \bar{u} \end{pmatrix} \begin{pmatrix} \bar{s} \\ \bar{c} \end{pmatrix} \begin{pmatrix} \bar{b} \\ \bar{t} \end{pmatrix}$$

proton: uud $u \bar{u} d \bar{d} s \bar{s} c \bar{c} \dots$

$\Rightarrow$ gluons in the sea

gluon: mediator of

strong interaction (Quantum Chromodynamics)



Tevatron: CERN

$$p \longrightarrow \bar{p}$$

$$\sqrt{s} = 1.96 \text{ TeV}$$

Symm. beam collider

$$E_{p,\bar{p}} \approx 1 \text{ TeV}$$

HERA CERN

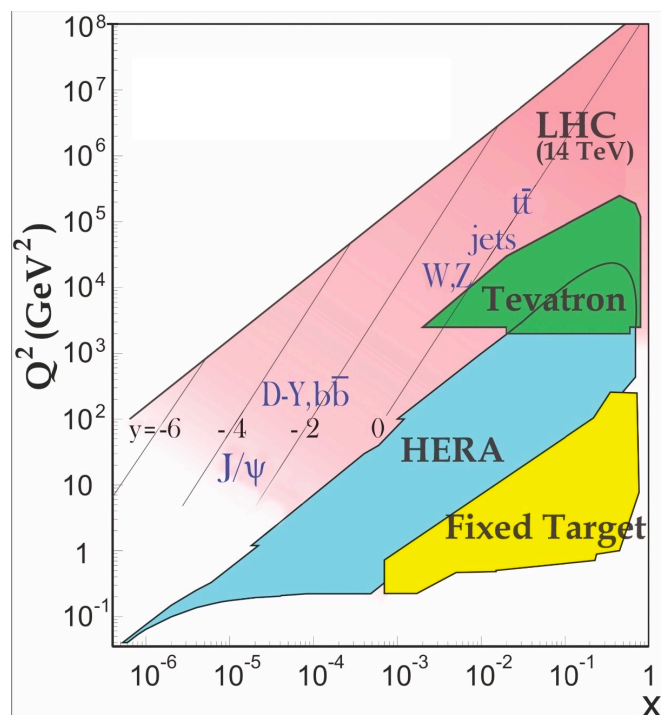
$$p \longrightarrow e^-$$

$$E = 920 \text{ GeV}$$

$$E_e = 27 \text{ GeV}$$

LHC CERN

$$\sqrt{s} \approx 13.6 \text{ TeV}$$



$\Rightarrow$ measure cross section $\Rightarrow$ fit to data for

$\Rightarrow$ extract $f_i(x)$ for F_1, F_2 protons/quarks.

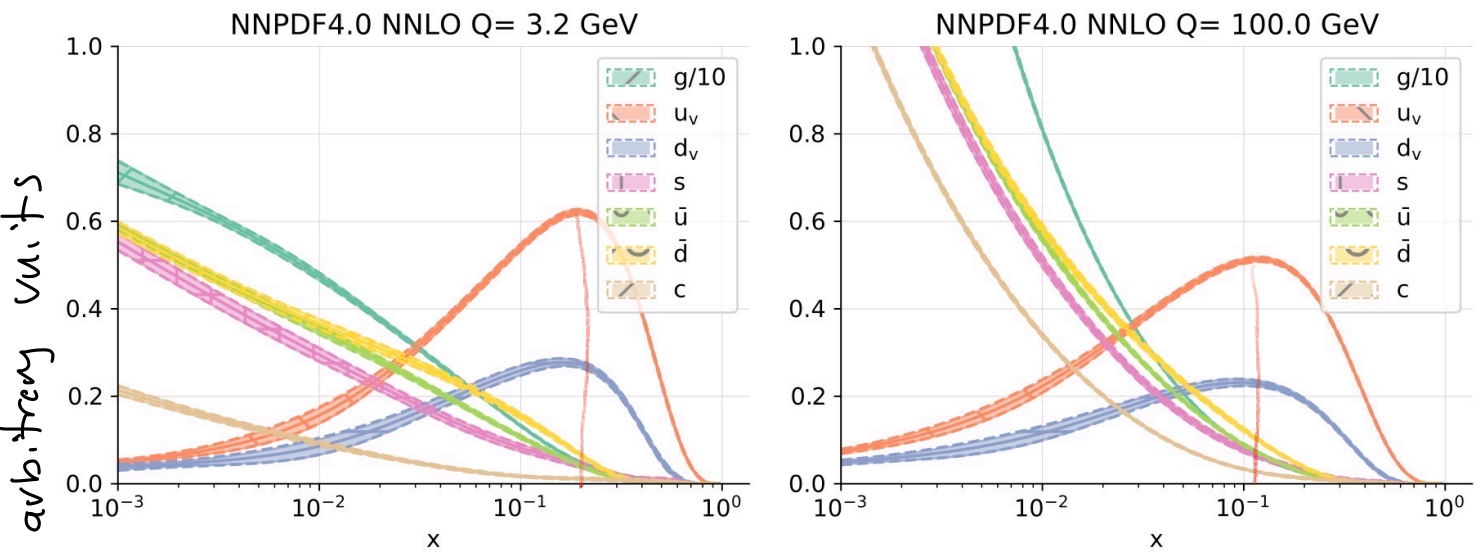


Figure 1.1. The NNPDF4.0 NNLO PDFs at $Q = 3.2$ GeV (left) and $Q = 10^2$ GeV (right).

u_v : valence up quark

$\bar{u}$: sea $\bar{u}$ anti-quark

$$x = \frac{|P_{\text{parton}}|}{|P_{\text{nucleon}}|}$$

$f_i \equiv u_v, d_v, \bar{u}, \bar{s}$
in these plots

$$\sum_i P_i = P_{\text{nucleon}} \Rightarrow \int_0^1 dx \left[\sum_i f_i(x) \right] x = 1$$

$$= \int_0^1 dx \left[\sum_{\text{flavors}} q_f(x) + \bar{q}_f(x) + g(x) \right] x = 1$$

Momentum sum rule

proton = uud

$$\int_0^1 dx [u(x) - \bar{u}(x)] = 2$$

2 valence up quarks.

$$\int_0^1 dx [d(x) - \bar{d}(x)] = 1$$

1 valence down quark

Experimental
proof for pions $\Rightarrow$

quarks? what is the
relation.

1950-60, $p, n, \pi^\pm, \pi^0, K^0, K^\pm, \Delta(1232)$
 $\swarrow$ $\underbrace{\hspace{1cm}}_{140 \text{ MeV}}$ $\underbrace{\hspace{1cm}}_{500 \text{ MeV}}$ 1.232 GeV
 $\underbrace{\hspace{1cm}}_{1 \text{ GeV}}$

Gell-Mann
Zweig

proposed quarks to explain the
mass structure of baryons and mesons

Baryon: $q_1 q_2 q_3$

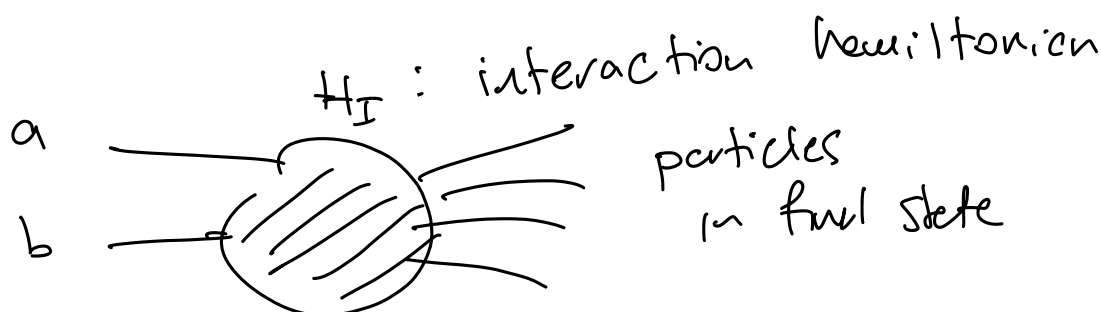
anti-baryon: $\bar{q}_1 \bar{q}_2 \bar{q}_3$

Meson: $q_1 \bar{q}_2$

anti-meson: $\bar{q}_1 q_2$

tetraquarks $q_1 q_2 q_3 q_4$

Symmetry to explain pattern of particle
masses.



#events $\propto |M|^2 \rho(E)$

$$\mu = \langle f | H_I | i \rangle$$

To study $H_I \Rightarrow$ look at symmetries/
selection rules.

Symmetry: transformation acts on a state
leaves the state unchanged

T : transformation
 $| \psi \rangle = \text{state}$

$\psi = \psi(\underline{q}, \underline{x}, t)$
quantum numbers: spin

First Nöther's theorem:

symmetry $\Leftrightarrow$ conserved quantity / charge

Internal Symm. : act on internal quantum numbers

External Symm. : act on $\vec{x}, t$

External Symm.:

Continuous :

translation of $\vec{x}$ $\rightarrow$ $\vec{p}$ conserved

rotation of $\vec{x}$ $\rightarrow$ $\vec{L} = \vec{r} \times \vec{p}$

translation of t $\rightarrow$ E

Discrete Symm.:

Parity : $P \psi(\vec{x}, t) = \psi(-\vec{x}, t)$

Time reversal : $\pi \psi(\vec{x}, t) = \psi(\vec{x}, -t)$

Internal Symm.:

Discrete :

Charge conjugation $\mathcal{C}$: charge electric charge

Continuous :

Isospin

Isospin internal Cont. Sym

Heisenberg n 940 MeV $\frac{\Delta m}{m} \approx 1/1000$
 p 939 MeV

very similar masses

most nuclear scattering experiments $\sigma(x+p) \approx \sigma(x+n)$

proposed nucleon = $\begin{pmatrix} p \\ n \end{pmatrix}$ doublet of Isospin. I

Isospin has the same algebra as spin/angular mom.

$$|p\rangle = |I=1/2, I_3=+1/2\rangle \quad |n\rangle = |I=1/2, I_3=-1/2\rangle$$

Similarly for pions:

$$\pi + p \rightarrow ?$$

π^+, π^- : identical mass.

$$m_{\pi^0} \approx m_{\pi^\pm}$$

$$\frac{\Delta m}{m} \approx \frac{5}{140}$$

$\pi^\pm, \pi^0$ same mass within 5%

=> isospin triplet

$$|\pi^+\rangle = |I=1, I_3=+1\rangle$$

$$|\pi^0\rangle = |1, 0\rangle$$

$$|\pi^-\rangle = |1, -1\rangle$$

Experimental evidence

$$|p\rangle = |1/2, 1/2\rangle$$

$$|n\rangle = |1/2, -1/2\rangle$$

deuterium:



$$I = 0, 1$$

$$I_3 = 0$$

$$\} |0, 0\rangle \text{ ? Singlet}$$

$$\} \begin{matrix} |1, 0\rangle \text{ ?} \\ |1, +1\rangle \\ |1, -1\rangle \end{matrix} \text{ triplet}$$

what if deuterium is a triplet

$|1, 1\rangle$ pp not observed

$|1, 0\rangle =$ deuterium $(p)(n)$ $|p\rangle|n\rangle$

$|1, -1\rangle$ nn not observed

$\Rightarrow$ hypothesis $|d\rangle \equiv |0, 0\rangle$ isospin singlet